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Localization Algorithm

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Introduction

Problem Definition I

Ground robots require accurate localization to navigate and map environments autonomously. This work evaluates the suitability of:

- **Visual SLAM algorithms** (e.g., ORB-SLAM2, ORB-SLAM3), and
- **IRSNet-based CNN approaches** for acoustic localization.

The platform under consideration uses:

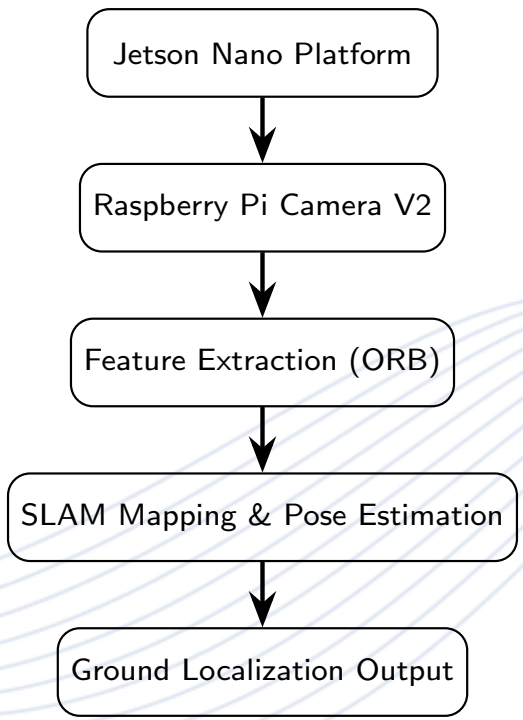
- **NVIDIA Jetson Nano Developer Kit** — embedded GPU for edge AI tasks.
- **Raspberry Pi Camera Module V2 (Sony IMX219, 8 MP)** — RGB input for visual odometry.

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Hardware Setup

System Overview I

- Jetson Nano provides GPU acceleration for real-time SLAM.
- Raspberry Pi Camera V2 provides monocular RGB images.
- Lightweight robotics middleware (e.g., ROS Noetic) for data flow.



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SLAM-Based Localization

Feature-Based Visual SLAM I

ORB-SLAM2 and **ORB-SLAM3** are state-of-the-art real-time visual SLAM systems.

- Detect and track ORB features across frames.
- Estimate camera pose via bundle adjustment.
- Map 3D landmarks for loop closure and re-localization.

Performance on Jetson Nano has been validated by Krško et al. (2025)Krško, Čeh und Jelušič, 2025, demonstrating reliable real-time localization with the Raspberry Pi Camera V2.

- Mur-Artal und Tardós, 2017 — ORB-SLAM2 core algorithm.
- Campos u. a., 2021 — ORB-SLAM3 visual-inertial extension.

Active SLAM Approaches I

Active SLAM algorithms plan robot motion to reduce map uncertainty while localizing. Ahmed et al. (2023)Ahmed u. a., 2023 review key advances integrating exploration and mapping. However, computational complexity makes them less suitable for real-time embedded deployment on Jetson Nano without simplification.

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IRSNet-Based CNN Localization

IRSNet Overview I

IRSNetSun, Fu und Teng, 2024 is a convolutional neural network for *acoustic* source localization. It estimates sound source positions using spectrogram features and improved residual structures. **Limitation for**

Ground Localization:

- Requires microphone arrays, not camera data.
- Lacks geometric consistency for visual odometry.
- Heavy inference cost on Jetson Nano compared to ORB-SLAM.

Thus, IRSNet is not optimal for vision-based robot localization.

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Comparative Evaluation

Algorithm Comparison I

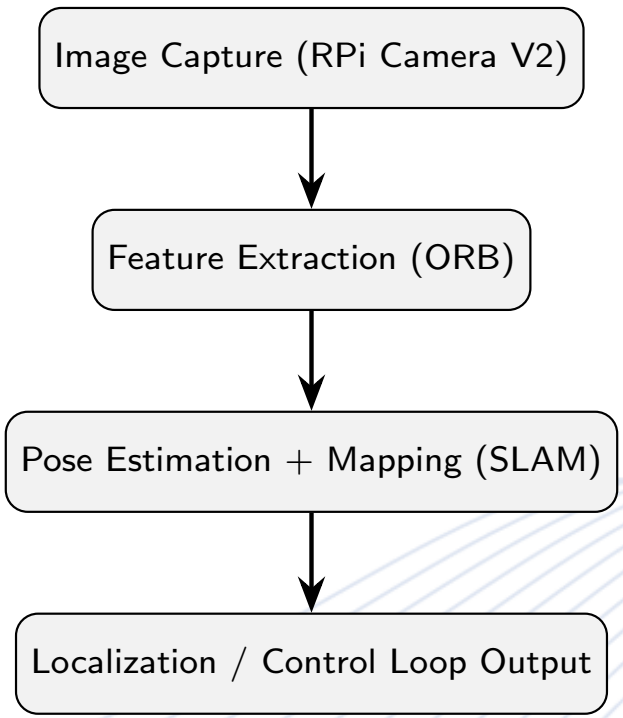
Method	Sensor	Accuracy	Jetson Nano Suitability
ORB-SLAM2 Mur-Artal und Tardós, 2017	Camera (RGB)	High (cm-level)	Excellent (20–30 fps)
ORB-SLAM3 Campos u. a., 2021	Camera + IMU	Very High / Robust	Good (15 fps)
Active SLAM Ahmed u. a., 2023	Camera + Depth	High / Exploratory	Moderate (CPU intensive)
IRSNet Sun, Fu und Teng, 2024	Microphone Array	Acoustic only	Not applicable

Tabelle: Comparison of Localization Methods for Jetson Nano + Raspberry Pi Camera V2.

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Flowchart

Ground Localization Pipeline I



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Conclusion

Summary and Recommendation I

- **IRSNet** excels in acoustic tasks, but not visual localization.
- **ORB-SLAM2 / ORB-SLAM3** are proven, efficient, and open-source for vision-based ground robots.
- Optimizations for Jetson Nano: reduce frame resolution, enable loop closure at intervals, use GPU-accelerated features.
- Future work: integrate lightweight CNN depth estimation with ORB-SLAM3 for enhanced floor-plane awareness.

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References

Key References I

- Mur-Artal R. and Tardós J.D., "ORB-SLAM2," IEEE TRO (2017)Mur-Artal und Tardós, 2017
- Campos C. et al., "ORB-SLAM3," IEEE TRO (2021)Campos u. a., 2021
- Ahmed M.F. et al., "Active SLAM: A Review," Sensors (2023)Ahmed u. a., 2023
- Sun D. et al., "IRSNet," Remote Sensing (2024)Sun, Fu und Teng, 2024
- Krško J. et al., "Evaluation of ORB-SLAM2 on Jetson Nano," Sensors (2025)Krško, Čeh und Jelušič, 2025

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Vielen Dank

für Ihre Aufmerksamkeit

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Quellen

References I

- Ahmed, Muhammad Farhan u. a. (2023). "Active SLAM: A Review on Last Decade". In: *Sensors* 23.19. Comprehensive review of Active SLAM techniques., S. 8097. DOI: 10.3390/s23198097.
- Campos, Carlos u. a. (2021). "ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial, and Multi-Map SLAM". In: *IEEE Transactions on Robotics* 37.6. Improved version of ORB-SLAM with visual-inertial fusion and multi-map capabilities., S. 1874–1890. DOI: 10.1109/TRO.2021.3075644.
- Krško, Jaka, Luka Čeh und Andrej Jelušič (2025). "Evaluation of ORB-SLAM2 on NVIDIA Jetson Nano Using Monocular and Stereo Raspberry Pi Camera Modules". In: *Sensors* 25.4. Practical evaluation of ORB-SLAM2 performance on Jetson Nano with Raspberry Pi Camera V2., S. 1025. DOI: 10.3390/s25041025.

References II


Mur-Artal, Raul und Juan D. Tardós (2017). "ORB-SLAM2: An Open-Source SLAM System for Monocular, Stereo, and RGB-D Cameras". In: *IEEE Transactions on Robotics* 33.5. Fundamental feature-based visual SLAM system for monocular, stereo, and RGB-D sensors., S. 1255–1262. DOI: 10.1109/TRO.2017.2705103.


Sun, Dajun, Xiaoying Fu und Tingting Teng (2024). "A Deep Learning Localization Method for Acoustic Source via Improved Input Features and Network Structure". In: *Remote Sensing* 16.8. IRSNet CNN architecture for acoustic source localization (non-visual domain)., S. 1391. DOI: 10.3390/rs16081391.